FINC5001 FOUNDATION IN FINANCE

TOPIC 3 – DEBT AND EQUITY VALUATION

PRE-WORKSHOP QUESTIONS

Pre-workshop self-access questions set is designed for you to self-access your understanding of the lecture content. You are expected to complete these questions prior to the workshop. These questions will not be discussed in the workshop.

Pre-Q1

David Jones' is considering issuing bonds with ten years to maturity to match the expected life of its store transformation plans. The bonds would have a face value of $1,000 and an annual coupon of $80. Similar bonds currently trade at a YTM of 8% p.a. compounded annually. Calculate the expected issue price of a David Jones' bond. Is this trading at par, at a premium or at a discount? What would be the price if the annual coupon was set at $60 instead? Or $100 instead?

Pre-Q2

Estimate the value of the following shares:

a) A share in a corporation that has been paying a semi-annual dividend of $0.40 in recent years, has no growth in earnings and that has a level of risk requiring a return on equity of 12.50% p.a. compounding annually.

b) A share that has just paid an annual dividend of $1.35, with a dividend growth rate of 2% p.a. expected in perpetuity and a required rate of return of 13% p.a.

c) A share issued by a company with current earnings of $2 per share. These earnings are expected to grow at 4% p.a. forever but only 30% of earnings are paid out as annual dividends. Their cost of equity is 12% p.a. compounding semi-annually.

d) A share in a company whose ROE and payout ratio is expected to remain constant at 8% p.a. and 60% respectively forever. The required return on equity is also 8% p.a. annually compounded. The next annual dividend is expected to be $0.75 per share.

FINC5001

Question 1

David Jones' is considering issuing bonds with ten years to maturity to match the expected life of its store transformation plans. The bonds would have a face value of $1,000 and an annual coupon of $80. Similar bonds currently trade at a YTM of 8% p.a. compounded annually. Calculate the expected issue price of a David Jones' bond. Is this at par, at a premium or at a discount? What would be the price if the annual coupon was set at $60 instead? Or $100 instead?

If the **annual coupon amount is $80,** the coupon rate is:

$$Coupon\ rate = \frac{Coupon\ amount}{Face\ value} = \frac{80}{1000} = 0.08$$

As the coupon rate = YTM, then the price of the bond would be selling at par:

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n} = 80\left[\frac{1-(1+0.08)^{-10}}{0.08}\right] + \frac{1000}{(1+0.08)^{10}}$$

$$D = 536.8065 + 463.1935 = \$1{,}000.00$$

If the **annual coupon amount is $60** instead, the coupon rate is:

$$Coupon\ rate = \frac{Coupon\ amount}{Face\ value} = \frac{60}{1000} = 0.06$$

As the coupon rate < YTM, then the price of the bond would be selling at a discount:

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n} = 60\left[\frac{1-(1+0.08)^{-10}}{0.08}\right] + \frac{1000}{(1+0.08)^{10}}$$

$$D = 402.6049 + 463.1935 = \$865.80 < \$1000$$

If the **annual coupon amount is $100 instead**, the coupon rate is:

$$Coupon\ rate = \frac{Coupon\ amount}{Face\ value} = \frac{100}{1000} = 0.10$$

As the coupon rate > YTM, then the price of the bond would be selling at a premium:

$$D = C\left[\frac{1-(1+r_d)^{-n}}{r_d}\right] + \frac{F}{(1+r_d)^n} = 100\left[\frac{1-(1+0.08)^{-10}}{0.08}\right] + \frac{1000}{(1+0.08)^{10}}$$

$$D = 671.0081 + 463.1935 = \$1{,}134.20 > \$1000$$

Question 2

Estimate the value of the following shares:

a) **A share in a corporation that has been paying a semi-annual dividend of $0.40 in recent years, has no growth in earnings and that has a level of risk requiring a return on equity of 12.50% p.a. compounding annually.**

$$Effective\ SA\ \ rate = \left(1 + \frac{r_{nom}}{m}\right)^{mn} - 1 = \left(1 + \frac{0.125}{1}\right)^{1\times\frac{1}{2}} - 1 = 0.060660171$$

$$PV = \frac{D}{r_e} = \frac{0.40}{0.060660171} = \$6.59$$

b) **A share that has just paid an annual dividend of $1.35, with a dividend growth rate of 2% p.a. expected in perpetuity and a required rate of return of 13% p.a.**

$$D_1 = 1.35 \times 1.02 = \$1.377$$

$$PV = \frac{D_1}{r_e - g} = \frac{1.377}{0.13 - 0.02} = \$12.52$$

c) **A share issued by a company with current earnings of $2 per share. These earnings are expected to grow at 4% p.a. forever but only 30% of earnings are paid out as annual dividends. Their cost of equity is 12% p.a. compounding semi-annually.**

$$D_0 = 2 \times 0.3 = \$0.60$$

$$D_1 = 0.60 \times 1.04 = \$0.624$$

The frequency of the dividend payments and the compounding frequency of the cost of equity do not match. The cost of equity needs to be converted into an EAR:

$$Effective\ annual\ rate = \left(1 + \frac{r_{nom}}{m}\right)^{mn} - 1 = \left(1 + \frac{0.12}{2}\right)^{2\times 1} - 1 = 0.1236$$

$$PV = \frac{D_1}{r_e - g} = \frac{0.624}{0.1236 - 0.04} = \$7.46$$

d) **A share in a company whose ROE and payout ratio is expected to remain constant at 8% p.a. and 60% respectively forever. The required return on equity is also 8% p.a. annually compounded. The next annual dividend is expected to be $0.75 per share.**

$$b = plowback\ ratio = (1 - 0.6) = 0.4$$

$$g = b \times ROE = 0.4 \times 0.08 = 0.032$$

$$PV = \frac{D_1}{r_e - g} = \frac{0.75}{0.08 - 0.032} = \$15.63$$